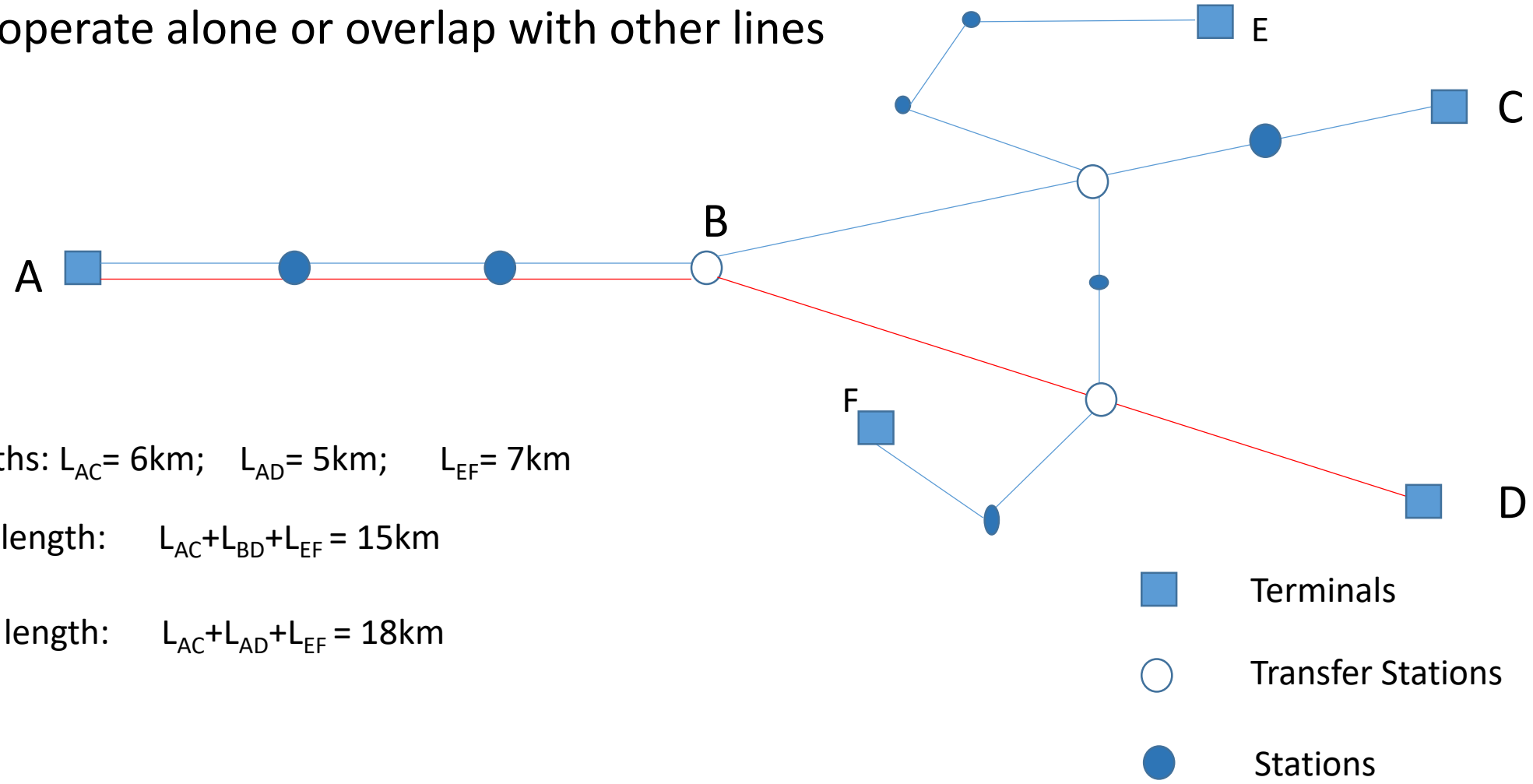
TRANSIT SYSTEM

- Transportation offered is the service
- Operations covers system management, scheduling and functioning from the operating agency's point of view

Line, Network, Stop and station

- **Transit Line** – Infrastructure and service on a fixed alignment with predetermined schedule
- **Transit Network –** A set of transit lines that connect with or cross each other and that are coordinated for efficient operation and provision of integrated services in an area

- **Line Length** - One-way distance between two terminals
- **Network length** - Total length of all alignments served by one or more lines
- **Total line (route) length** – Sum of all line lengths regardless of whether they operate alone or overlap with other lines



Line lengths: L_{AC}= 6km; L_{AD}= 5km; L_{EF}= 7km

Network length: $L_{AC}+L_{BD}+L_{EF}$ = 15km

Total line length: $L_{AC}+L_{AD}+L_{EF}$ = 18km

Transit (ROW) – Strip of land on which a transit line operates

With respect to the degree of separation from other vehicles and pedestrians, it is classified into – C, B & A categories

Category C – Surface street with mixed traffic
Category B – Longitudinally separated from other traffic; at-grade separation
Category A – Grade separated

Transit Stop – Location along a line at which transit vehicles stop to pick-up or drop off passengers
Transit Station – A special structure and facility for boarding/alighting , waiting and transfer
Terminals – End stations

Vehicles, Transit Units and Fleet size

Transit Unit (TU) – Set of n vehicles travelling physically coupled together
Fleet Size (N_f) = Total number of vehicles needed for operation of a line, or of an entire network

$$N_f = N + N_r + N_m$$

N = Vehicles required for regular service determined by the peak hour operation
N_r = Vehicles needed for reserve
N_m= Vehicles on maintenance or repair
Fleet Utilization Factor (ψ) – Percentage of fleet available for service

$$\psi = \frac{N + Nr}{N_f}$$

Headway and Frequency

Headway h – Time interval between the moments two successive TUs pass a fixed point on a transit line in the same direction

- Clock Headways – If headways longer than 6 min then desirable to use only values divisible into 60
- Policy headway (h_p)- Longest headway for accessible min. level of service
- Min. headway h_{min} = Max (h_{wmin} ; h_{smin})

Frequency f – Number of TUs passing a point on a transit line in one direction during one hour

$$f = \frac{60}{h}$$

f= TU/hr

h=min/TU

CAPACITY, WORK AND UTILIZATION

- Vehicle Capacity – Spaces per vehicle
- Transit Line Capacity – Spaces per hour

- Utilization Coefficient α also known as load factor – Ratio of utilized capacity to offered capacity

$$\alpha = \frac{P}{C}$$

P = prs/hr
C = sps/hr
α = Prs/sps

- Vehicle Capacity C_v – max. no. of spaces for passengers a vehicle can accommodate
- TU Capacity $C_{TU} = n \times C_v$

- Vehicle Line Capacity c = Max. no. of vehicles that can pass a fixed point on a line in one direction in one hour, regardless of single unit or in trains

$$c = n \times f_{max}$$

- Operating Time T_o: Scheduled time interval b/w departure of a TU from one terminal and its arrival at the other terminal on the line

$$T_o=\Sigma(t_{ri}+ t_{si})$$

- Terminal Time t_t= Time a TU spends at a line terminal

- Cycle Time (T)

$$T= T_o'+ T_o''+t_t'+t_t''$$
$$= 2(T_o+t_t)$$ (if both ways' terminal times & operating times are equal)

  - Cycle time is the basic time unit for scheduling transit service, and it strongly influences investment and operating costs

- **Deadhead time t_d** = The portion of TU travel time during which the TU is not in passenger (revenue) service (desirable to minimize it). It includes travel from depot to the line and back.

- Platform time T_p = Total time a TU is in operation

$$T_p =kT+2t_d$$

[Platform time when a TU makes k round trips on a line and has both deadhead times equal]

Passenger Travel times

- Access time t_a
- Waiting time t_w
- Transfer time t_f
- Origin & Destination Travel time T_{od}

$$T_{od} = t_a + t_w + t_o + t_f$$

SPEEDS

Table 1.3 Transit speeds and corresponding travel times

Category	Speed Designation	Symbol	Corresponding Travel Time
Vehicle speed	1. Actual vehicle	V, v	—
Alignment speeds	2. Maximum technical	V_{max}	—
	3. Line design	V_d	—
	4. Legal	V_ℓ	—
Vehicle-on-line speeds	5. Programmed	V_g	—
	6. Running	V_r	t_r
	7. Station-to-station	V_s	T_s
	8. Operating	V_o	T_o
	9. Cycle	V_c	T
Passenger speeds	10. Platform	V_p	T_p
	11. Access	V_a	t_a
	12. Travel on line	V_o	T_o
	13. Origin-destination	V_{od}	T_{od}

- **Actual vehicle speed** – Designated as V
- **Max. technical speed V_{max}** – Highest speed a transit vehicle is physically capable of achieving on a straight horizontal way under normal weather conditions
- **Line Design speed** – Max. speed transit vehicle can achieve on a given section of line with adequate comfort & safety

$$V_d = \frac{\Sigma S_i}{\Sigma \frac{S_i}{V_i}} = \frac{L}{\Sigma \frac{S_i}{V_i}}$$

V- km/h ; S,L - km

- **Legal Speed** – Max. speed transit vehicle can legally operate on a given section of line
- **Programmed speed** – Speed that transit vehicle can actually achieve: Automated train operations can operate it precisely

[None of these speeds include transit stops, i.e. not considering requirements to stop and related time delays]

Vehicle-on-Line Speeds

- **Running speed V_r**– Average speed TUs achieve from leaving one station (stop) to arriving at the next one

$$V_r = \frac{60Si}{t_{ri}}$$ V-km/h; S-km; t-min

- **Station to station speed V_s** – Average speed of travel between moments a TU leaves two adjacent stations; it includes running and station dwell time

$$V_{si} = \frac{60Si}{t_{ri} + tsi}$$

- **Operating Speed or Travel speed V_o** – The average speed of TU travel along transit line with j spacings (or on a section of it)

$$V_{si} = \frac{60L}{T_o} = \frac{120L}{T_o' + T_o''}$$

❖ *Based upon the defined Time, calculate its respective speed*

- **<u>Cycle Speed V_c-</u>** Average speed of a TU for a complete round trip on a line

$$V_c = \frac{60.\ 2L}{T} = \frac{120L}{T}$$

- **<u>Platform Speed Vp –</u>** Average speed of TUs operating on a line from the time they leave depot (garage or yard) until they return to it

$$V_p = \frac{120(k.L+Ld)}{k.\,T+2td} = \frac{60Lp}{T_p}$$

where L_d is deadhead distance;
L_p is the total distance the TU travels while absent from the depot